Práctica Dos: Sistema cardiovascular

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

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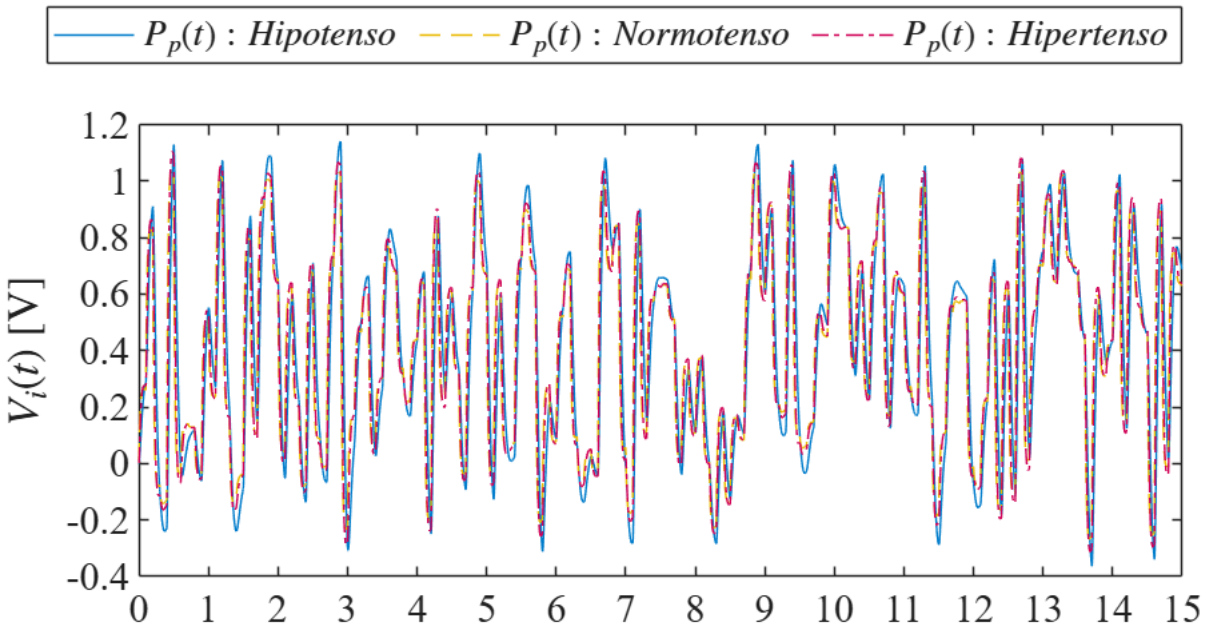
Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system('MSFP2NFGA.slx')
parameters.StopTime = '15';
parameters.Solver = 'ode15s';
```

```
parameters.MaxStep = '1e-2';
x = sim('MSFP2NFGA',parameters);
t_data = (0:0.1:15)';
t = (0: 0.001: 15)';
Pa = interp1(t_data,x.Pa,t, 'previous');
writematrix(Pa,'señales.xlsx')
```

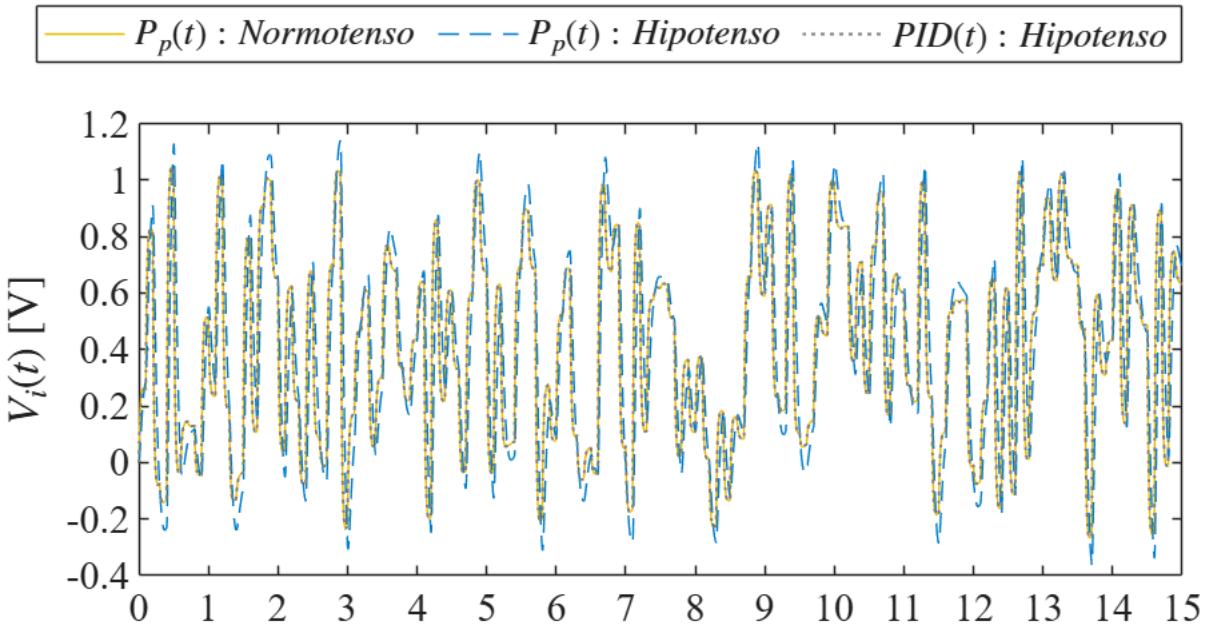
Respuesta Hipotenso, Normotenso e Hipertenso

```
plotsignals(x.t,x.PpL,x.PpN,x.PpH,'individuos')
```



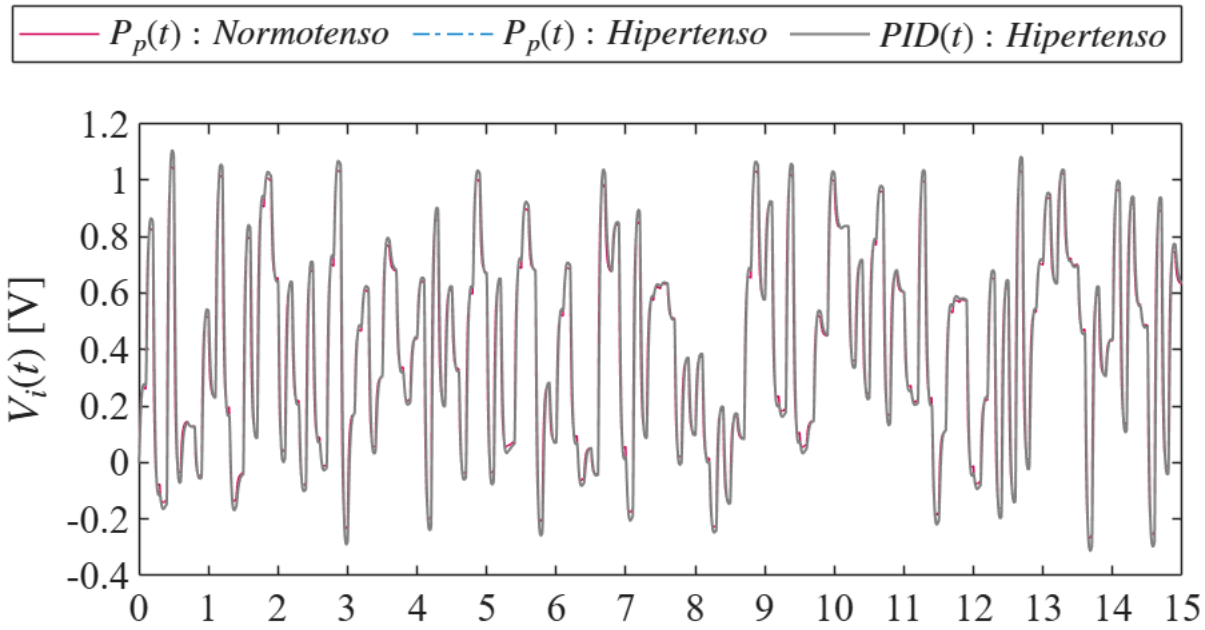
Respuesta Normotenso Hipotenso y Controlador

```
plotsignals(x.t,x.PpN,x.PpL,x.PIDL,'hipotenso')
```



Respuesta Normotenso, Hipertenso y Controlador

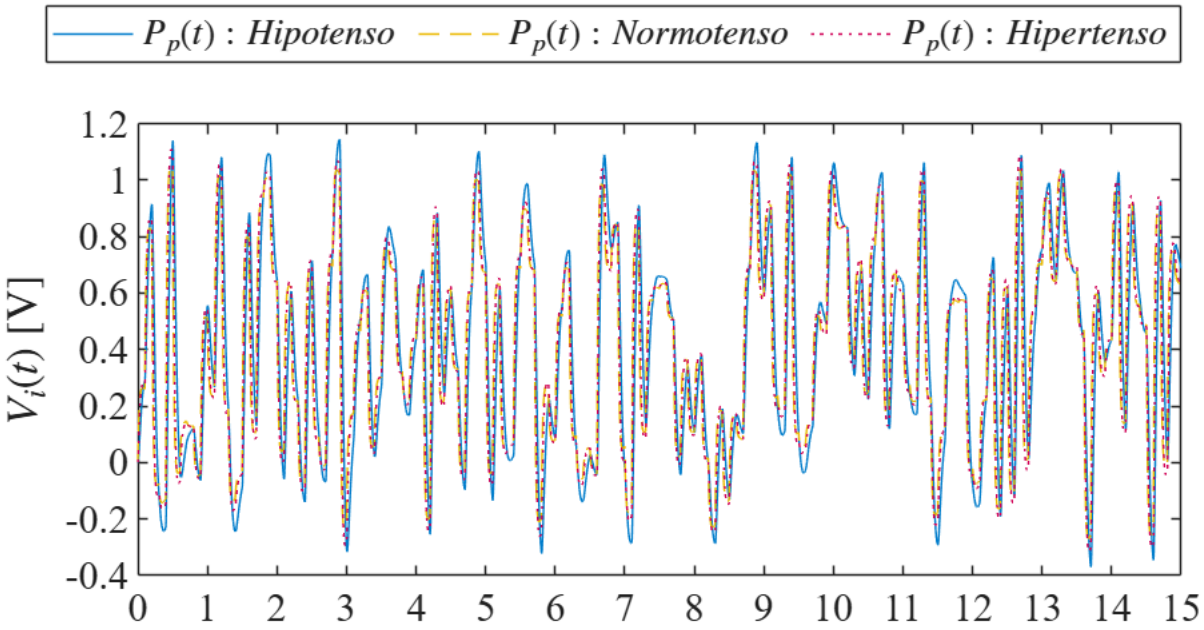
```
plotsignals(x.t,x.PpN,x.PpH,x.PIDH,'hipertenso')
```



Modelo de EOs

```
par_hipo =[0.01,0.001,1.00,3.500]; %[Z,L,R,C]
par_normo=[0.01,0.001,1.00,1.500];
par_hiper=[0.01,0.001,1.00,1.000];
PpN = cardio(par_normo,Pa,t);
PpL = cardio(par_hipo,PpN,t);
PpH = cardio(par_hiper,PpN,t);
```

```
plotsignals(t,PpL,PpN,PpH,'odes')
```




```
function Pp= cardio(par,Pa,t)
Z= par(1); L= par(2); R= par(3); C= par(4); dt=t(2);
n= round(t(end)/dt);
FL = zeros(length(t),1); FL(1)= 0;
Pp = zeros(length(t),1); Pp(1)= 0;
for i=1:n
    FL(i+1)=FL(i)+(-Pp(i)/L+Pa(i)/L)*dt;
    Pp(i+1)=Pp(i)+(FL(i)/C- (1/(Z*C)+1/(R*C))*Pp(i)+Pa(i)/(Z*C))*dt;
end
end
```

Funcion: Respuesta de la simulacion

```
function plotsignals(t,x1,x2,x3,caso)
set(figure(),'color','w')
set(gcf,'units','inches','position',[1,1,7,3.5])
set(gca,'fontname','times new roman','fontsize',15)
hold on; grid off; box on;
clr1 = [0, 125, 204]/255;
clr2 = [235,185,0]/255;
clr3 = [209,0,86]/255;
clr4= [0.5,0.5,0.5];
if caso == "individuos"
```

```
    plot(t,x1,'-','Linewidth',0.25,'Color',clr1)
    plot(t,x2,'--','Linewidth',0.25,'Color',clr2)
    plot(t,x3,'-.','Linewidth',0.25,'Color',clr3)
    L =
legend('$P_p(t):Hipotenso$','$P_p(t):Normotenso$','$P_p(t):Hipertenso$');
elseif caso=="hipotenso"
    plot(t,x1,'-','Linewidth',0.25,'Color',clr2)
    plot(t,x2,'--','Linewidth',0.25,'Color',clr1)
    plot(t,x3,':','Linewidth',1,'Color',clr4)
    L =
legend('$P_p(t):Normotenso$','$P_p(t):Hipotenso$','$PID(t):Hipotenso$');
elseif caso=="hipertenso"
    plot(t,x1,'-','Linewidth',0.25,'Color',clr3)
    plot(t,x2,'-.','Linewidth',0.25,'Color',clr1)
    plot(t,x3,'-','Linewidth',1,'Color',clr4)
    L =
legend('$P_p(t):Normotenso$','$P_p(t):Hipertenso$','$PID(t):Hipertenso$');
elseif caso == "odes"
    plot(t,x1,'-','Linewidth',0.25,'Color',clr1)
    plot(t,x2,'--','Linewidth',0.25,'Color',clr2)
    plot(t,x3,':','Linewidth',0.25,'Color',clr3)
    L =
legend('$P_p(t):Hipotenso$','$P_p(t):Normotenso$','$P_p(t):Hipertenso$');
end
set(L,'interpreter','latex','location','northoutside','orientation','horizont
al')
ylabel('$V_i (t)$ [V]','interpreter','latex')
xlim([0 15]); xticks(0:1:15)
ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)
exportgraphics(gcf,[caso,'_matlab.pdf'],'ContentType','vector')
end
```